

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination

April 2018

EC767 EMBEDDED SYSTEM DESIGN

Date: 05.04.2018, Thursday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 Differentiate Waterfall life cycle model and Spiral life cycle model. **[07]**

Q - 2 Attempt any two: **[14]**

- (a) Write an embedded C program for AVR microcontroller to send “CHARUSAT IS BEST” serially with baud rate of 9600. (assume XTAL= 11.0592MHz)
- (b) Interface two common anode seven segment displays with ATMEGA32. Develop an embedded C code to display UP counter on seven segments.
- (c) Interface MAX7221 SPI slave with ATMEGA32. Develop an embedded C program to display 00 to 99 on seven displays at interval of one second.

Q - 3 Attempt any two: **[14]**

- (a) What is Inter-integrated Circuit (I2C)? With the help of bit pattern, Explain I2C start, I2C stop, I2C read and I2C write frame.
- (b) Interface 16x2 Alphanumeric LCD with Atmega32 and Write a C code to display “CHARUSAT” on it.
- (c) Write an embedded C program to generate 5kHz square wave on PB7 pin using timer 0 in normal mode AVR.

SECTION – II

Q - 4 Define an embedded System. Draw block diagram of typical embedded system. **[07]**
With help of suitable example explain it.

Q - 5 Attempt any two: **[14]**

- (a) Differentiate following :
 - i) Microcontroller and microprocessor
 - ii) RISC and CISC
- (b) Explain software co-design issues. With neat sketch explain FSM for automatic tea/coffee vending machine & seat belt warning system.

- (c) Explain various kinds of memory which are required to develop the embedded system.

Q – 6 (a) What is sensor? Illustrate its role in embedded system design with help of suitable example. **[07]**

Q – 6 (b) What is Serial Peripheral Interface (SPI)? Differentiate I2C and SPI protocol. **[07]**